

Update - 29/07

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Supervisor: Professor Maurizio Corbetta

Point of Discussion

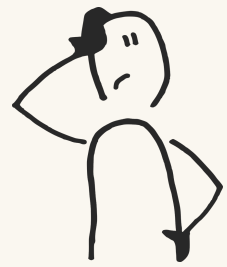
- **Progress Report**
 - Deadline: 20 August
 - Presentation: 17-18 September
 - Reviewers to Find
- **Thesis**
 - Nemesis?
- **Period abroad**
 - Deco's Lab → Nemesis, Contact with Jackub
- **Projects**
 - MAIN FOCUS: Nemesis
 - Collaboration - Personal Training

A Feature Space of Stroke Disease

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Rationale



Problem

In Stroke, we have several separated stories:

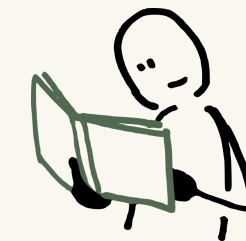
- Anatomical (**lesion**)
- Structural disconnections (**SDCs**)
- Functional features (**fMRI**)
- Physiological features (**EEG**)
- Phenotypic (demographics + clinical/behavioral **scores**)

→ different data types, different scales,



Idea

To get a holistic view, we need to bring these signals together, not simply stack them side by side, but test whether they share the same **latent structure** and if they possibly describe the same underlying **deficit**.



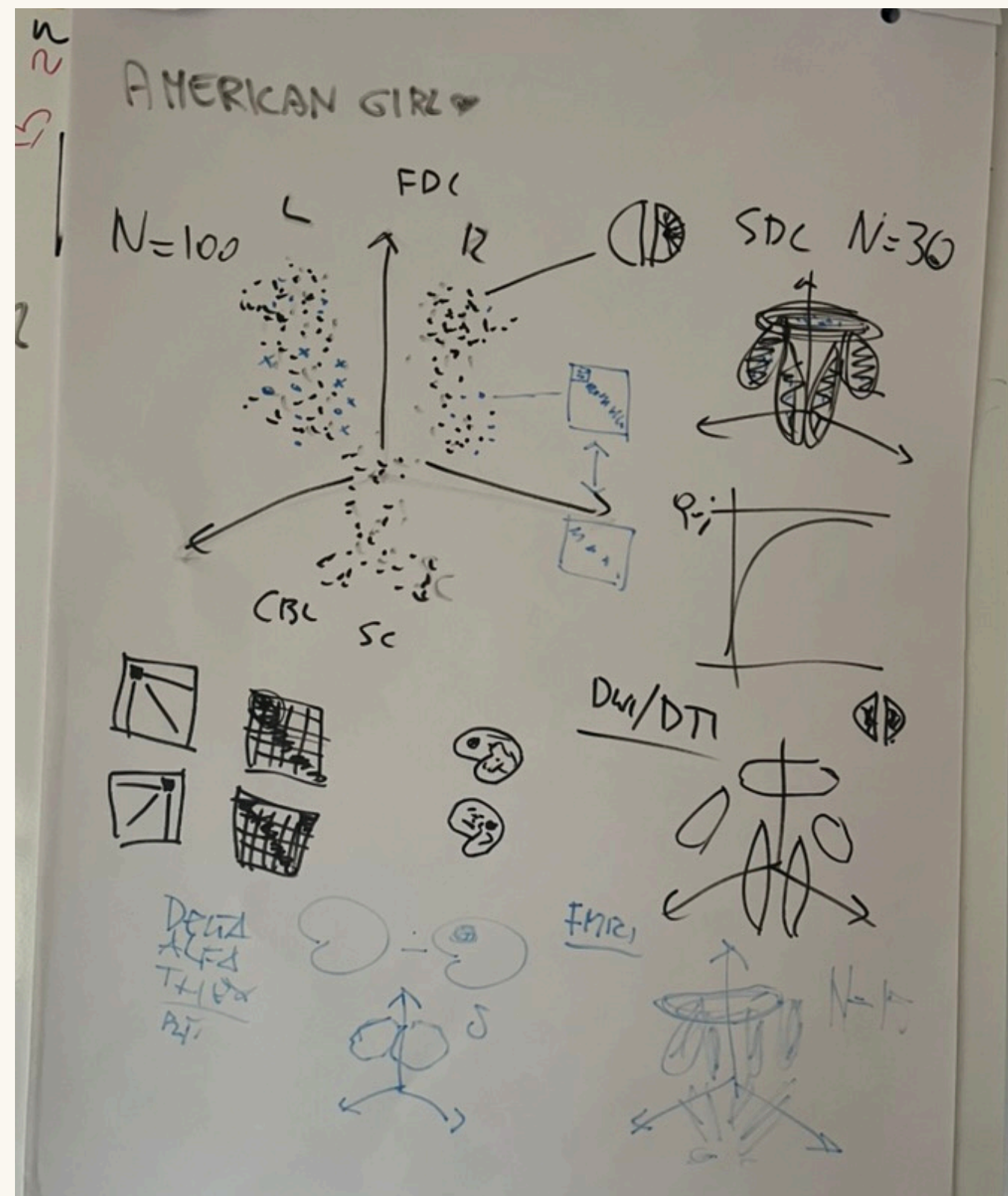
Methodology

We want to build a **multimodal feature space**:

- reduce each modality to a low-dimensional embedding,
- cluster each embedding independently,
- then test whether these clusters overlap across modalities and whether they are jointly correlated with behavioral outcomes.

Lieterature

- **Corbetta et al., 2015** - Common Behavioral Clusters and Subcortical Anatomy in Stroke
- **Siegel et al., 2016** - Disruptions of network connectivity predict impairment in multiple behavioral domains after stroke
- **Siegel et al., 2018** - Re-emergence of modular brain networks in stroke recovery
- **Griffis et al., 2019** - Structural Disconnections Explain Brain Network Dysfunction after Stroke
- **Bonkhoff et al., 2020** - Acute ischaemic stroke alters the brain's preference for distinct dynamic connectivity states
- **Griffis et al., 2020** - Damage to the shortest structural paths between brain regions is associated with disruptions of the resting-state
- **Salvalaggio et al., 2020** - Post-stroke deficit prediction from lesion and indirect structural and functional disconnection
- **Thiebaut de Schotten et al., 2020** - Brain disconnections link structural connectivity with function and behavior
- **Bisogno et al., 2021** - A low-dimensional structure of neurological impairment in stroke
- **Talozzi et al., 2023** - Latent disconnectome prediction of long-term cognitive-behavioral symptoms in stroke
- **Idesis et al., 2023** - A low-dimensional embedding of brain dynamics enhances diagnostic accuracy and behavioral prediction in stroke
- **Facchini et al., 2023** - A common low-dimensional structure of cognitive impairment in stroke and brain tumors
- **Idesis et al., 2024** - Whole-brain model replicates sleep-like slow-wave dynamics generated by stroke lesions
- **Bisogno et al., 2025** - Large-scale network topography of stroke predicts functional outcome after mechanical thrombectomy
- **Pini et al., 2026** - Longitudinal Degeneration of Microstructural and Structural Connectivity Patterns Following Stroke



**From
Brainstorming to
Ideas and First
Pipelines →**

Workflow until now

Dataset

- Explored and mapped the shared clinical dataset structure of EBRAIN data.
- The source data already follows the **BIDS standard**, and aligned our internal data model to it.
- Scoped to 4 target cohorts
 - **Wahsu**
 - **PSP**
 - **PASPORT**
 - **UKLFR** - Friburgo
- Data restricted to derivatives
 - **Lesion masks** and functional connectivity data
 - Both co-registered to MNI152NLin6Asym space
 - Functional connectivity computed across **12 atlas** combinations

Lesions Masks

Dataset	N
UKLFR	697
WashU	202
PSP	168
PASPORT	83
Total	1150

FC WashU

Category	N
Lesion mask	202
Functional connectivity (FC)	225
HC (healthy controls)	56
ST (stroke patients)	169
Intersection (lesion $\cap$ FC)	169
Lesion present, FC missing	33
FC present, lesion missing	56

Workflow until now

Ideas

Stroke research today produces disconnected stories: anatomical, structural, functional, physiological → **how do these modalities overlap?**

Low-dimensional embedding

- Embed each modality independently, then cluster
 - Lesions masks → where the damage is (topography)
 - SDC matrices → which pathways are disrupted (structural disconnection profile)
- **Lesion clusters** are expected to largely recover vascular territories - topography is anatomically constrained.
- **SDC is a many-to-one mapping**: different lesion topographies can produce the same disconnected pathway → subjects in different lesion clusters may converge onto a single SDC cluster
- The SDC embedding should show a more compressed structure (fewer effective clusters / lower intrinsic dimensionality) than the lesion embedding (*Thiebaut de Schotten et al., 2020*)

Workflow until now

Ideas

Compare clusterings across modalities

- **Contingency table crossing** the lesion-based clusters with the SDC-based clusters
 - *Patients who were grouped together based on their lesion, do they still end up in the same SDC cluster, or do they scatter across several?*
 - Where and how the groups diverge
- **Cross-cluster convergence**
 - *Patients with clearly different lesions may still share the same disconnected pathway, so we check whether distinct lesion clusters collapse onto a single SDC cluster.*
- **Metrics** to check how much the two clusters agree overall
- ...

Workflow until now

Ideas

Clinical interpretation

- Characterize each cluster with a mean lesion map and mean disconnectome map, then relate clusters to clinical scores (NIHSS, behavioral domains)
- *Siegel et al. (2016)* showed that sensorimotor deficits are better explained by lesion topography, while associative/memory deficits are better explained by functional (dis)connectivity

Workflow until now

Pipelines

1. Data Engineering

- Data Retrieval
- Lesion matrix: voxel-wise binary matrix ($1150 \times 254,865$ voxels), all sites resampled to a common MNI grid
- FC matrix
 - masked by lesion extent before vectorization (reproduces *Siegel et al. 2016's*)
 - computed across 12 atlas combinations (Yan $\times$ Tian $\times$ Buckner)
- Exploratory data analysis on the resulting matrices and metadata
 - per-dataset coverage, demographics, missing-value patterns, lesion laterality, correlation across clinical/behavioral scores
- Both outputs land as ready-to-use 2D matrices, same interface for every downstream ML step

Workflow until now

Pipelines

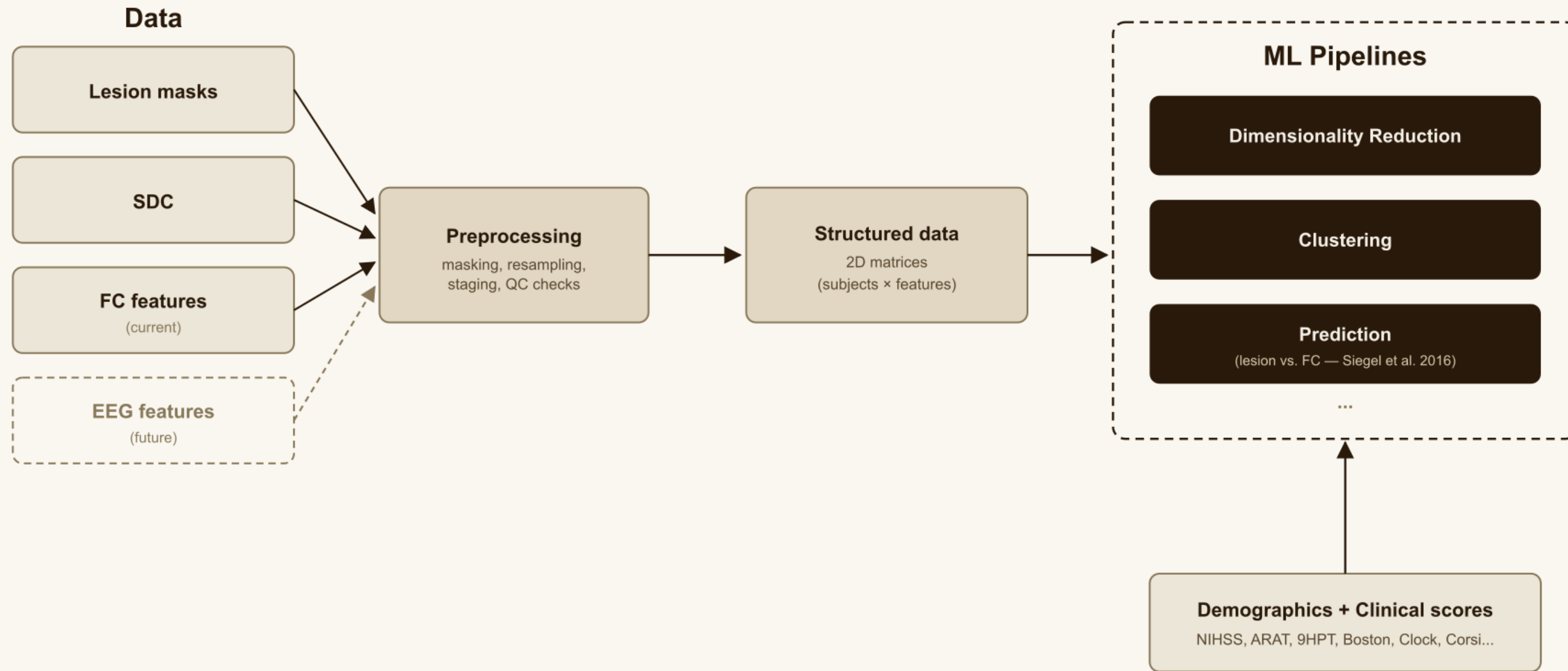
2. Structural Disconnection - in progress

- BCBToolKit (*Foulon, Thiebaut de Schotten*) orchestrated end-to-end: staging → Stage 1 (disconnectome) → Stage 2 (parcellated features), with per-subject validation at each step

3. ML and Statistical Pipeline

- Dimensionality reduction:
 - UMAP, t-SNE, PaCMAP, PCA
- Clustering:
 - k-means, agglomerative, GMM, spectral, DBSCAN, swept across every embedding
- Ongoing hyperparameter tuning to select the best-performing configuration per method before promoting values to production.
- Siegel, 2016 → **TO DO**

Pipeline Structure - Coding Level



Workflow until now

Ideas, Hints, Sugestions?